

Andrew Stewart

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PROFESSIONAL SUMMARY

Analytical, research-driven engineer with a Master of Science in Industrial and Systems Engineering, concentrating in human factors & ergonomics. Experienced in experimental design, human subjects research, and ergonomic workplace assessment, including a thesis examining the effects of augmented reality on postural balance during simulated ladder-based workplace tasks; findings have been accepted for publication in a peer-reviewed journal.

EDUCATION

Master of Science — Industrial & Systems Engineering, Human Factors & Ergonomics Concentration

Virginia Polytechnic Institute and State University, Blacksburg, VA, August 2024 – May 2026, GPA: 3.77

Thesis — Effects of Performing Augmented Reality Tasks on Postural Sway While on a Ladder

Findings accepted for publication in a peer-reviewed journal (forthcoming)

- Examined AR's effects on postural balance for construction workers and inspectors performing tasks on ladders — an interaction with no prior formal research
- Designed a study where participants completed four simulated ladder-based work tasks with AR assistance; data collected and analyzed in MATLAB

Bachelor of Science — Industrial & Systems Engineering, Human-Computer Interaction Minor

Virginia Polytechnic Institute and State University, Blacksburg, VA, August 2020 – May 2024, Overall GPA: 3.31, In-Major GPA: 3.56

PROJECT/RESEARCH EXPERIENCE

Quality of Life Plus Student Engineer, Interdisciplinary Design Course @ VT *Fall 2023 – Spring 2024*

- Collaborated on a senior design project with Quality of Life Plus, a nonprofit serving veterans and first-responders with injuries or disabilities
- Designed and built a lower limb exoskeleton enabling the client to walk comfortably and maintain active lifestyle
- Managed the team's budget and oversaw purchasing of components
- Oversaw safety considerations throughout exoskeleton development

Research Assistant, Madigan Biomechanics Group *Spring 2024*

- Investigated slip recovery training as a tool for workplace injury prevention
- Conducted tracking analysis on experiment recordings and managed subsequent database entry
- Supported participants through human-subjects testing procedures

Research Assistant, Mind Music Machine Lab *Spring 2023*

- Designed novel XR interactions for a 3D planetary model exhibit ("Large Scale Object Manipulation in XR")
- Surveyed literature on virtual reality and interactive arts to inform design decisions
- Designed and tested new 3D VR interaction methods through a user experiment
- Analyzed data using statistical tools and presented findings

SKILLS

- **Human factors & ergonomics:** workplace assessment, biomechanical analysis, injury prevention research
- **Human subjects experimental research:** IRB-compliant protocols, behavioral and social research methods
- **Human-centered design & HCI:** UI/UX principles, usability evaluation, AR/XR interaction design
- **Data collection, management, & analysis:** Excel data management; MATLAB data collection and analysis
- **Communications:** oral presentations, live experimental demos, corresponding with stakeholders

AFFILIATIONS

Cognitive Engineering for Novel Technologies (COGENT) Lab (2025–Present); Madigan Biomechanics Group (2024–Present); Human Factors and Ergonomics Society (2024–Present); Mind Music Machine Lab (2023)

LEADERSHIP

Senior Staff Assistant, The Marching Virginians (2025); Vice President of Membership, Kappa Kappa Psi (2023); Section Leader, The Marching Virginians (2023–2025); Public Relations Officer, The Marching Virginians (2022); Rank Captain, The Marching Virginians (2021–2022)

QUALIFICATIONS

Order of the Engineer (2024); CITI Social and Behavioral Research Certified (2022, 2025); CITI Basic Responsible Conduct of Research Certified (2024)